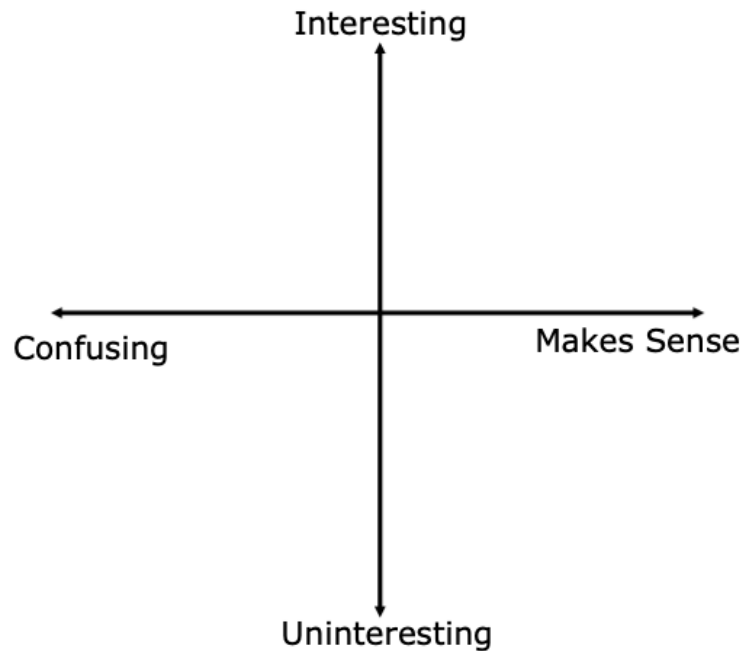




2.5.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants to indicate how you feel about today's learning target, "I can add and subtract polynomials."



2. Describe why you plotted your point in that location.



Unit 2, Lesson 6: Multiplying Binomials With Algebra Tiles



Benchmark

MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.



Learning Target

- ☐ I can multiply binomials using algebra tiles.



2.6.1 Warm-Up: Math Puzzle

A math puzzle is shown with one value missing.

3	4	6	2
5	2	3	5
2	6	?	3
1	7	5	7

- Find the missing number in the puzzle.
- Describe the pattern used to find the missing number.



2.6.2 Exploration: Multiplying With the Area Model

Work with your partner to complete the following.

- Three multiplication problems are set up using an area model.
 - Complete the area models to find each product.

Model A

	20	+ 5
10		
+5		

Model A's Product

Model B

	20	+ 7
20		
-5		

Model B's Product

Model C

	20	-2
12		
+3		

Model C's Product

- b. Match each area model to the multiplication expression it represents.

Multiplication Expression	Area Model
$(15)(27)$	☐ Model A ☐ Model B ☐ Model C
$(15)(25)$	☐ Model A ☐ Model B ☐ Model C
$(15)(18)$	☐ Model A ☐ Model B ☐ Model C

- c. Complete the statements.

In the area models, each rectangle represents part of

the ☐ factor.
☐ product. Parts of each ☐ factor
☐ product are

multiplied to find four partial products. The product of

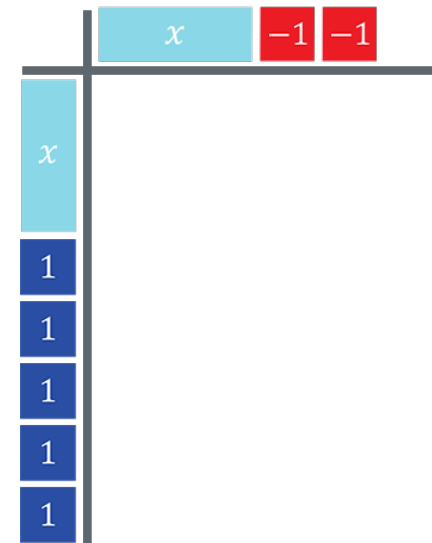
the two factors is found by ☐ adding
☐ multiplying the partial products.



2.6.3 Guided Instruction: Multiplying Binomials

Similar to the area model, algebra tiles can be used to model the product of two binomials.

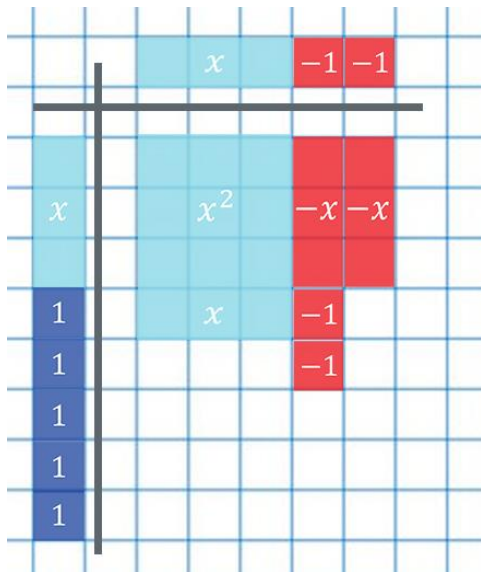
1. Ty wants to find the product of $(x + 5)$ and $(x - 2)$. He uses algebra tiles to model multiplying $(x + 5)(x - 2)$.



Similar to the area model, the factors are placed along the width and the height of a rectangular area.

- a. Discuss with your partner how each factor is represented using algebra tiles. Write a summary of your discussion.

- b. Complete Ty's model to find the product of $(x + 5)$ and $(x - 2)$ by drawing and labeling the missing tiles.



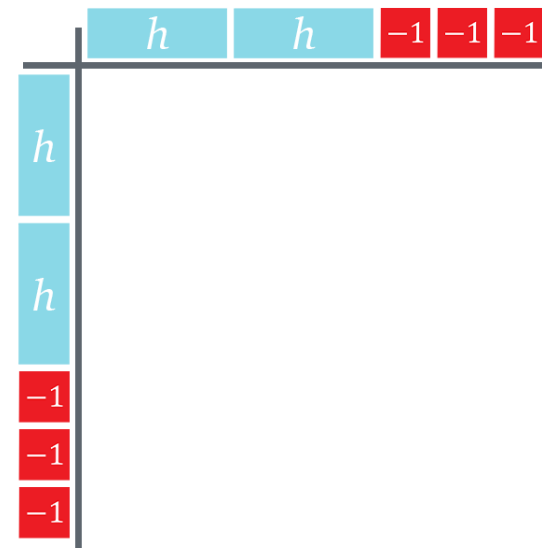
- c. Determine how many of each tile are represented in the product and write the numbers in the table.

x^2 tiles	$+x$ tiles	$-x$ tiles	$+1$ tiles	-1 tiles

- d. Complete the expression that represents the product of $(x + 5)$ and $(x - 2)$ using the algebra tiles.

$$(x + 5)(x - 2) = \underline{\hspace{1cm}}x^2 + \underline{\hspace{1cm}}x - \underline{\hspace{1cm}}$$

2. Monica is using algebra tiles to find the product of $(2h - 3)^2$.



- a. How do the algebra tiles model represent $(2h - 3)^2$?
- b. Complete Monica's model.
- c. What shape did the product of $(2h - 3)^2$ form using the algebra tiles?

Remember when a value is “squared,” the value is being multiplied by itself.

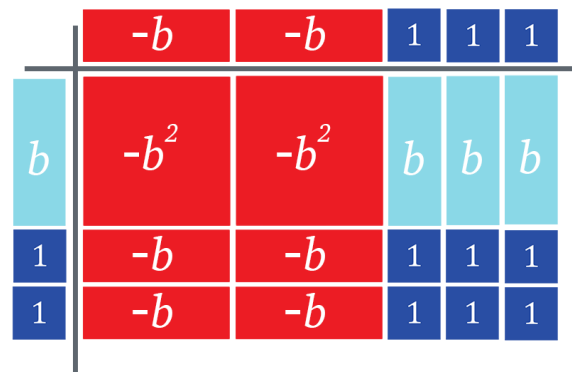
- d. Complete the table by writing how many of each tile are in the model of the product of $(2h - 3)^2$.

h^2 tiles	$+ h$ tiles	$- h$ tiles	$+ 1$ tiles	$- 1$ tiles

- e. Write the algebraic expression that represents the product of $(2h - 3)^2$.

Work with your partner to complete the following.

3. Annalee used algebra tiles to find the product of $(b + 2)(3 - 2b)$. Her model is shown.



- a. Complete the statements.

The factor $(b + 2)$ is represented along the

- ☐ width
☐ height

of Annalee’s model, while the factor

$(3 - 2b)$ is represented along the

- ☐ width.
☐ height.

- b. Annalee wrote the product is $2b^2 - 7b + 6$. Describe the error(s) Annalee made.

- c. Complete the statement.

The product of $(b + 2)(3 - 2b)$ is _____.



2.6.4 Guided Practice: Multiplying Binomials

1. Use algebra tiles to find the product of $(1 - r)(4r - 1)$.



$$(1 - r)(4r - 1) = \underline{\hspace{2cm}}.$$

2. Find the product of $(4x + 1)^2$ using algebra tiles.



$$(4x + 1)^2 = \underline{\hspace{2cm}}$$



2.6.5 Your Turn!

1. Use algebra tiles to find the product of $(2x - 1)(x + 6)$.



$$(2x - 1)(x + 6) = \underline{\hspace{2cm}}$$

2. Use algebra tiles to find the product of $(x - 4)(x + 4)$.



$$(x - 4)(x + 4) = \underline{\hspace{2cm}}$$



2.6.6 Wrap-Up: Cool Down

1. Use algebra tiles to find the product of $(x - 5)^2$.



$$(x - 5)^2 = \underline{\hspace{2cm}}$$



Unit 2, Lesson 7: Multiplying Binomials and Polynomials



Benchmark

MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.



Learning Targets

- ☐ I can multiply binomials.
- ☐ I can multiply polynomials.